

Inference Perf: A Benchmarking Tool for GenAI Inference

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Summary

Inference Perf is a generative AI (GenAI) inference performance benchmarking tool aimed at benchmarking and analyzing the performance of inference deployments. It is designed to be model server-agnostic, allowing for apples-to-apples comparisons across different model servers and serving stacks. It was created as a part of the inference benchmarking and metrics standardization effort in the Kubernetes wg-serving working group ([Kubernetes Authors, 2024](#)), and seeks to standardize tooling and metrics for measuring inference performance across the Kubernetes and model server communities.

Statement of need

With the rapid adoption of Large Language Models (LLMs) and GenAI, there is a growing need to accurately measure and compare the performance of inference serving systems. Different model servers (e.g., vLLM, TGI, SGLang) and deployment orchestrators (e.g., Kubernetes) introduce substantial variability in performance. Existing tools often lack standardized metrics or GenAI inference-specific capabilities (e.g., Grafana Labs (2017); Heyman et al. (2011)) or are tightly coupled to specific frameworks (e.g., vLLM Team (2023); Hugging Face (2023b); NVIDIA (2024)) where their goal is to provide a tool for developers working on the specific framework to benchmark their system. As a result, it is often hard to reproduce benchmark results across different serving stacks and environments. Inference Perf addresses this gap by providing a scalable, agnostic, and comprehensive benchmarking suite for GenAI workloads. It supports various real-world and synthetic datasets, different load patterns (e.g., burst, saturation), and integrates with standard cloud-native observability tools like Prometheus ([Prometheus Authors, 2012](#)), allowing it to benchmark both smaller-scale systems in development as well as large production-scale deployments orchestrated by Kubernetes. Crucially, it provides a standardized comparison between different model servers and serving stacks across various use cases.

State of the field

There are two kinds of performance benchmarking tools for GenAI inference that are commonly used to measure inference performance of a model serving stack:

1. Web-based benchmarks (e.g., Grafana Labs (2017); Heyman et al. (2011))
2. Model server benchmarks (e.g., vLLM Team (2023); Hugging Face (2023b); SGLang Team (2024); NVIDIA (2024))

Web-based benchmarks are generic web server benchmarking tools which offer battle-tested way to reliably generate traffic against specific HTTP endpoints. While these can be used to benchmark LLMs and GenAI workloads, they lack the standardized set of metrics that we

39 want to measure with inference, often at token level. To measure these token-level metrics,
40 streaming request support, tokenizer support and other features specific to the GenAI workload
41 that is being tested are needed. While some of these tools allow extensions, writing custom
42 extensions is restrictive in general and is not ideal for GenAI benchmarking.

43 Model server benchmarks are geared towards developers of the model server to repeatedly
44 measure performance improvements that are being made to that model server. While these
45 work well for benchmarking GenAI inference, they are very specific to the model servers and
46 don't work well for production workloads where different traffic patterns that simulate real-world
47 workloads are needed, especially to validate autoscaling, load balancing and intelligent routing,
48 which are staple features of these production systems.

49 There are also other tools like MLPerf (Reddi et al., 2020) which focus on competitive
50 hardware accelerator performance measurements by providing a standard load generation
51 tool and leaderboard for comparing inference performance across different accelerator chips.
52 However, they are not designed to benchmark production-scale workloads under various traffic
53 patterns and use cases.

54 The main contribution of Inference Perf is to provide a standardized model server-agnostic
55 tool that is designed to benchmark production-scale GenAI workloads for various real-world
56 use cases.

57 Software design

58 Inference Perf is built with a modular architecture comprised of several key components:

- 59 ▪ **DataGenerator:** Aligns prompt and generation lengths with user input, supporting fixed-
60 or variable-length tests for use cases like chat completion and summarization, including
61 both real-world and synthetic datasets.
- 62 ▪ **Load Generator:** Generates traffic patterns such as fixed RPS, bursts, or Poisson
63 distributions. It supports multi-process generation for high concurrency, which is a
64 critical requirement for benchmarking production-scale systems.
- 65 ▪ **Client:** Abstractions for different model servers, ensuring the tool can be extended to
66 support new model servers and protocols. Furthermore, the tool provides native support
67 for the industry-standard OpenAI API, enabling it to benchmark any compatible model
68 server using their chat and completion endpoints without necessitating modifications.
- 69 ▪ **Metrics/Data Collector:** Measures key performance indicators including time to first
70 token (TTFT), time per output token (TPOT), inter-token latency (ITL) and various
71 throughput metrics. It also supports exporting metrics to Prometheus, which can be
72 used to visualize metrics using tools like Grafana.
- 73 ▪ **Report Generator:** Produces detailed JSON reports with all the metrics collected during
74 benchmarking.
- 75 ▪ **Analyzer:** Analyzes the collected metrics and provides insights into the performance of
76 the model server by generating various charts and graphs.

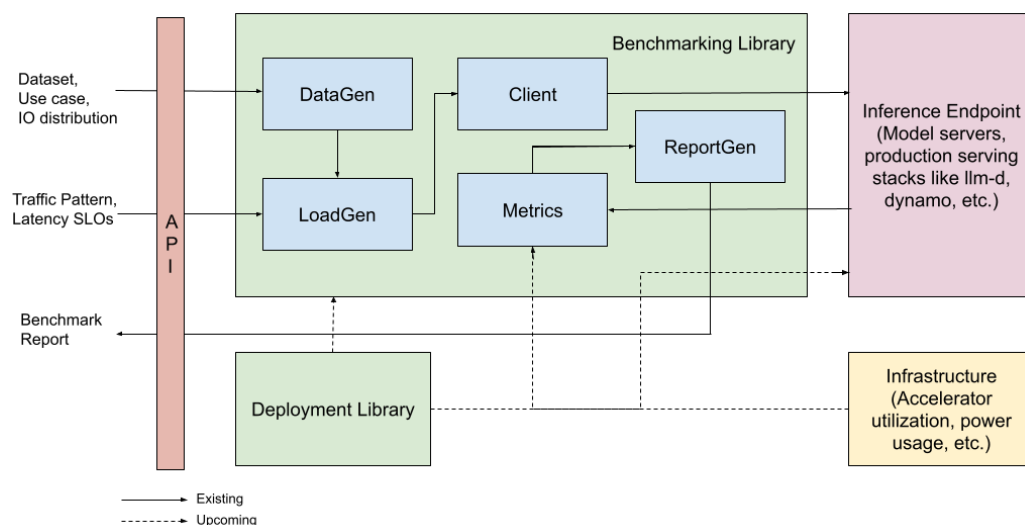


Figure 1: Architecture Diagram

Key features

- Scalability to support large production deployments with request rate generation up to 10k+ requests per second via a novel multi-process load generator capable of maintaining accurate QPS over longer durations.
- Support for multiple backends including vLLM (Kwon et al., 2023), SGLang (Zheng et al., 2024), and HuggingFace TGI (Hugging Face, 2023a). It is also extensible to support any serving stack which follows the OpenAI API, like llm-d (llm-d Team, 2025b) and NVIDIA Dynamo (NVIDIA, 2025) which are not model servers, but entire optimized inference stacks with advanced orchestration capabilities.
- Simulation of complex scenarios like multi-turn chat conversations, shared prefix caching and autoscaling.
- Comprehensive metrics collection from both the benchmarking client and the model server to aid in debugging performance issues and discrepancies.
- Observability into the load generated by the benchmarking client. This is important because benchmarking clients can be artificially constrained by external factors like resource contention on client machines, underlying Python library limitations, etc. which can lead to performance differences. Being able to observe these limitations is essential.
- Ability to replay traces to mimic production traffic using traces recorded from production and to reproduce the same load pattern in different runs.

Standardized metrics

Inference Perf defines the key metrics required to measure inference performance and aims to standardize these metrics and their definitions. The set of metrics measured by Inference Perf, as listed below, provides a comprehensive view of the performance of the inference server in terms of throughput and latency. Detailed definitions of the below metrics can be found in Inference Perf Contributors (2025b).

Throughput

- Output tokens/second
- Input tokens/second
- Requests/second

106 Latency

- 107 ■ Time per request (e2e Request Latency)
- 108 ■ Time to first token (TTFT)
- 109 ■ Time per output token (TPOT)
- 110 ■ Normalized time per output token (NTPOT)

111 Price-performance

112 The price-performance metrics below are not directly reported by Inference Perf but can be
113 computed from the metrics reported by the tool using the cost of the underlying hardware.
114 Inference Perf Contributors (2025b) provides the formulas to compute these additional metrics.

- 115 ■ Price per million output tokens
- 116 ■ Price per million input tokens
- 117 ■ Throughput per dollar

118 The above metrics can also be plotted into charts using the “analyze” command in the tool
119 at various request rates (QPS) to understand how the latency and throughput scales with
120 the load. The charts below demonstrate benchmarks run on the llama3.1-8b model with
121 NVIDIA H100 GPUs on vLLM version v0.11.0. Further details of the benchmark run and other
122 similar examples can be found in the demos section of the source repository (Inference Perf
123 Contributors, 2025c).

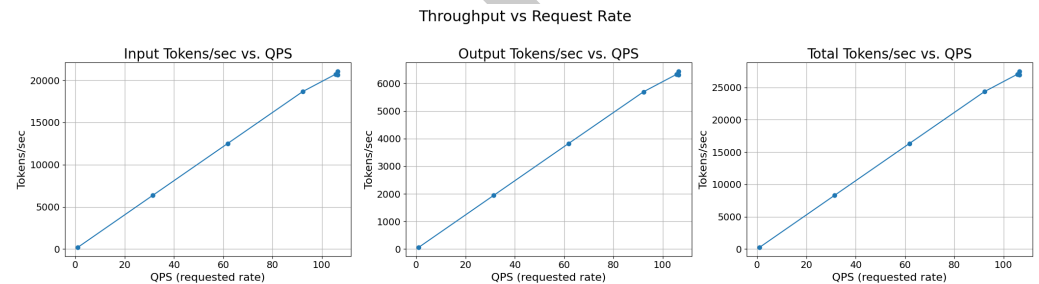


Figure 2: Throughput vs QPS

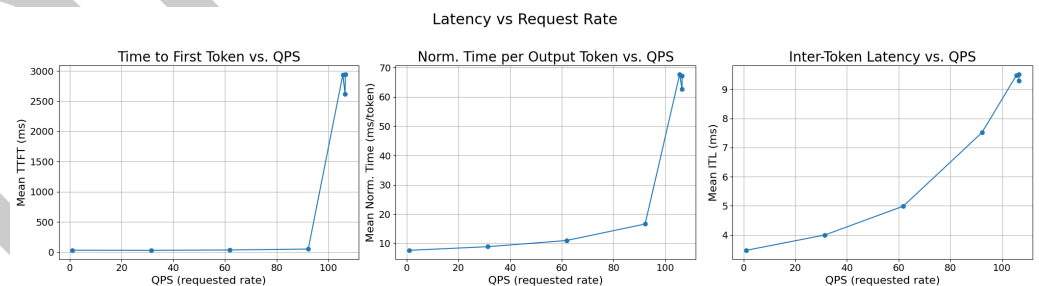


Figure 3: Latency vs QPS

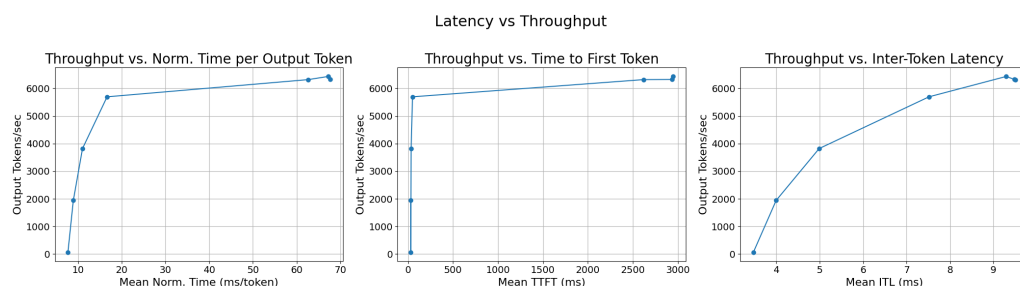


Figure 4: Throughput vs Latency

Research impact statement

Inference Perf is the primary benchmarking tool used by state-of-the-art open-source distributed inference frameworks like llm-d, which implements novel model serving optimizations for LLM orchestration such as multi-host serving, efficient load balancing and autoscaling, as seen in llm-d Team (2025b). An example usage of Inference Perf to benchmark and optimize can be found in llm-d Team (2025a). Inference Perf is also used by the Kubernetes community and different organizations for various evaluation and development purposes, as seen from the contributors and issue creators on GitHub (Inference Perf Contributors, 2025a).

AI usage disclosure

Our AI usage follows the Linux Foundation's guidance on AI usage (Linux Foundation, 2023) where contributors are allowed to use code-assist tools. All of the pull requests are manually reviewed and approved by at least two reviewers or maintainers of the project, and the contributor is responsible for the code quality and addressing any comments from the reviews. Since there are many contributors to this project, not every specific tool used by the contributors could be called out here.

Acknowledgments

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